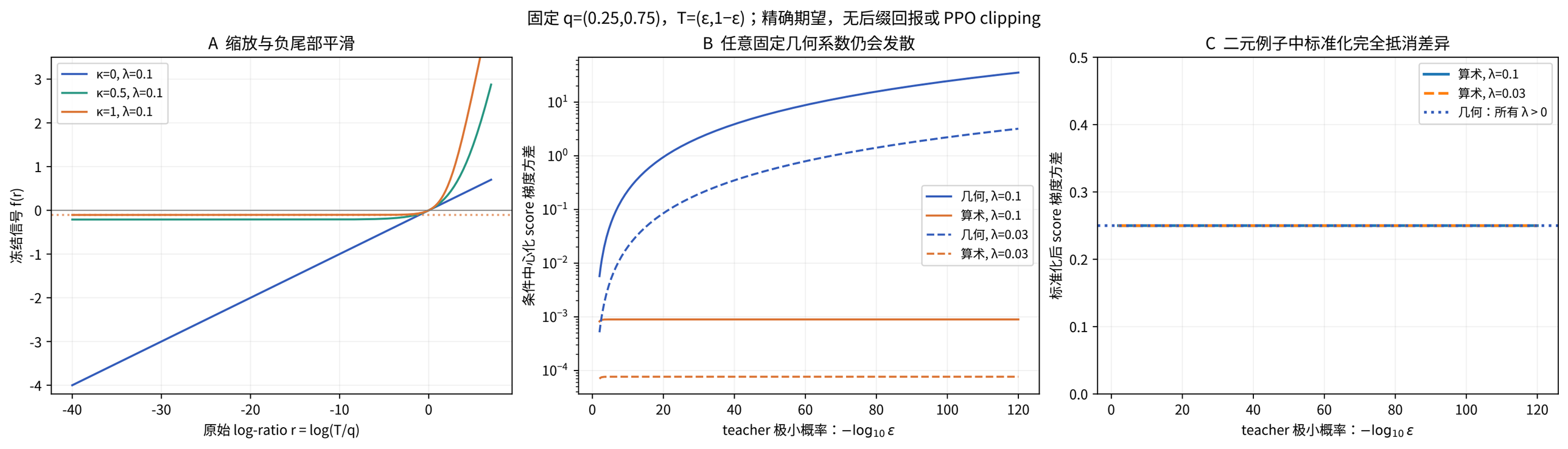
固定 q=(0.25,0.75)，T=(ε,1−ε)；精确期望，无后缀回报或 PPO clipping
A 缩放与负尾部平滑
κ=0, λ=0.1
κ=0.5, λ=0.1
κ=1, λ=0.1
冻结信号 f(r)
原始 log-ratio r = log(T/q)
B 任意固定几何系数仍会发散
几何, λ=0.1
算术, λ=0.1
几何, λ=0.03
算术, λ=0.03
条件中心化 score 梯度方差
teacher 极小概率：−log10 ε
C 二元例子中标准化完全抵消差异
算术, λ=0.1
算术, λ=0.03
几何：所有 λ > 0
标准化后 score 梯度方差
teacher 极小概率：−log10 ε